

OpenAI Math Digest

CATALOGUE 051

Kobayashi's canonical- ampleness conjecture

Catalogue guide & one proof overview

Independent reading guide · English edition

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Contents

The catalogue guide is followed by the article. The contents and citations are clickable. Wide tables are arranged as entries for print; the claims and source references are retained.

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[Catalogue 051 · Guide](#)

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CATALOGUE 051

Kobayashi's canonical-ampleness conjecture

This catalogue concerns canonical ampleness for compact hyperbolic Kähler manifolds. The official list contains one manuscript. The article presents its claims and proof route, with the scope of source checking stated at the end.

The single paper in official catalogue order

These summaries give the principal scope; consult the cited original results for the complete hypotheses and statements.

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01

Manuscript and overview

Canonical ampleness of compact hyperbolic Kähler manifolds →

Short name: Canonical ampleness

2026-09-23 · 31 pages

Original paper (GitHub) ↗

Principal claim

The canonical bundle of a compact connected Brody-hyperbolic Kähler manifold is ample. Consequences give generation for every $m \geq n+2$ and a finite étale product cover under a semialgebraic universal-cover hypothesis.

Theorem 1.1 · p. 1; Corollaries 10.1–10.2 · pp. 26–27 ↗

Role and available overview

Boundary-regular disc variations yield nefness and a volume bound; Monge–Ampère, Morse, projectivity and cone inputs complete the proof. The corollaries use separate inputs from 038 and 058.

Read the proof overview →

Entering the proof and its two consequences

Start with the [main proof overview](#), then follow [disc variations and the Hessian](#), [nefness](#), the [uniform volume bound](#), and [positive intersection](#). The overview also explains the subsequent steps through [bigness](#) and [projectivity](#) to [ampleness](#).

After the main theorem, the [all-exponents generation result](#) uses Fujita freeness from 038, while the [finite étale product cover](#) uses universal-cover classification from 058. These inputs do not enter the proof of the main theorem.

Sources and checking scope

The official subject is **Algebraic and complex geometry**. The reference catalogue is dated October 6, 2026 and pinned to commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`.

[OpenAI Research Catalog · Catalogue 051 · p. 7](#)

The manuscript is by OpenAI, dated September 23, 2026, and has 31 pages including references.

[Theorem 1.1 · p. 1](#); [Corollaries 10.1–10.2 · pp. 26–27](#)

The principal reductions in the proof and the statements and application hypotheses of twelve external references were compared. This does not independently verify every boundary-regularity, functional-analytic or smoothing estimate, re-prove the external theorems, or exhaust all dependencies. See the [article’s closing section](#) for the checking scope and unresolved items.

051 / 01

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CATALOGUE 051 · PAPER 01 / 1 Overview

Canonical ampleness of compact hyperbolic Kähler manifolds

From boundary-regular disc variations to canonical ampleness

The manuscript constructs extremal discs from the derivative bound supplied by compact hyperbolicity. Two uses of the same Hessian identity yield nefness and positive canonical volume. Projectivity is obtained before the final algebraic argument. This account follows the authors' argument and distinguishes checking the statements and uses of inputs from independently verifying complete proofs.

Manuscript author: OpenAI · 2026-09-23 · 31 pages, including references

[Read the manuscript on GitHub ↗](#)

1. Main results

Theorem 1.1 — Canonical ampleness

Let X be a compact connected Kähler manifold of complex dimension $n \geq 1$. If every holomorphic map $\mathbb{C} \rightarrow X$ is constant, then K_X is ample. In particular, the global sections of a positive tensor power $K_X^{\otimes m}$ define a holomorphic embedding of X into a complex projective space.

Theorem 1.1 · p. 1

Corollary 10.1 — All pluricanonical exponents

Let X be a compact connected Brody-hyperbolic Kähler manifold of complex dimension $n \geq 1$. For every integer $m \geq n + 2$, the bundle $K_X^{\otimes m}$ is globally generated.

Corollary 10.1 · p. 26

Corollary 10.2 — A product after a finite cover

Let X be a compact connected Brody-hyperbolic Kähler manifold of positive complex dimension. Assume that its ordinary universal cover is biholomorphic to a semialgebraic open subset of a complex projective variety, with openness in the ordinary complex topology. Then X admits a connected finite étale cover biholomorphic to

$$(D/\Gamma_0) \times F.$$

Here D is a bounded symmetric domain, Γ_0 is a discrete group of biholomorphisms acting freely, properly discontinuously and cocompactly on D , and F is a simply connected compact Brody-hyperbolic projective manifold. Either D or F may be a point.

Corollary 10.2 · pp. 26–27

References on the arrows

Unprefixed theorem, lemma, section and equation numbers refer to [Canonical ampleness](#). External inputs use the following keys.

- [PR](#) Barbara Drinovec Drnovšek; Franc Forstnerič — Poletsky–Rosay envelopes: Plurisubharmonicity of the envelope on the punctured determinant bundle, used in Proposition 6.2. *Indiana Univ. Math. J.* 61 (2012), 1407–1423; published version.
- [MA](#) Robert J. Berman — Monge–Ampère existence: Aubin–Yau existence in the stated arbitrary-volume-form setting; Theorem 8.1 and Proposition 8.2. *Math. Z.* 291 (2019), 365–394; repository copy with cover sheet.
- [HM](#) Jean-Pierre Demailly — Holomorphic Morse inequalities: Section growth and the Moishezon criterion in Lemma 9.1. *Ann. Inst. Fourier* 35(4) (1985), 189–229; Numdam scan with cover sheet.
- [MP](#) Yoshinori Namikawa — Moishezon projectivity: Projectivity of the smooth Kähler Moishezon manifold, used in Theorem 9.2. *arXiv:math/0101019v6*, 2001-11-12.
- [DT](#) Simone Diverio; Stefano Trapani — Canonical ampleness criterion: A comparison statement for Proposition 9.3, which the manuscript proves through the cone theorem; the curvature main theorem is not an input. *arXiv:1606.01381v2* (margin: 2016-08-04; manuscript date: 2019-04-19); author version.
- [LC](#) Osamu Fujino — Log cone and ampleness criteria: The klt perturbation, rational curve, and nef-plus-ample steps of Proposition 9.3. *Publ. RIMS* 47 (2011), 727–789; published version.
- [FF](#) OpenAI — Fujita freeness: Input only to Corollary 10.1, with $L=K_X$ and exponent $m-1$. 2026-09-23; fixed catalogue 038 manuscript.
- [SC](#) OpenAI — Semialgebraic universal covers: Input only to Corollary 10.2: product classification of the ordinary universal cover. 2026-09-24; fixed catalogue 058 manuscript.

- [KA Shoshichi Kobayashi](#) — Finite automorphisms: Finiteness of $\text{Aut}(F)$ from ample K_F in Corollary 10.2. Tohoku Math. J. (2) 11 (1959), 184–190; published version.
- [CG Alexandre Sukhov; Alexander Tumanov](#) — Cauchy–Green operator: The $L^p \rightarrow W^{1,p}$ right inverse used in Lemma 4.1 and Proposition 4.3. arXiv:1411.3989v4, 2016-04-05.
- [IF Helge Glöckner](#) — Banach implicit functions: Construction of the centered holomorphic family in Proposition 4.3. arXiv:math/0303320v1, 2003-03-25.
- [OK Franc Forstnerič; Jasna Prezelj](#) — Oka principle: The holomorphic frame over a Stein disc in Lemma 4.1, via its $\text{GL}_n(\mathbb{C})$ frame bundle. arXiv:math/0305238v1, 2003-05-16; first page unnumbered.

Each citation gives the result and pages. When printed pagination differs from the PDF position, both are shown. GitHub previews do not automatically jump to the cited page.

2. Theorem 1.1 — Proof overview of the main theorem

Proof target : [Theorem 1.1](#) · p. 1

Proof route. The [Hessian identity](#) branches into [nefness](#) and a [volume bound](#), which meet at [positive top intersection](#). Bigness, projectivity and ampleness follow in that order. The external inputs to the two corollaries do not enter this proof.

Fix a background Kähler metric k and set $R_h = -i\partial\bar{\partial} \log \det h$ and $d(f) = |f'(0)|_k^2$. On the disc $\mathbb{D}$, use

$$A_q(f) = \frac{1}{\pi} \int_{\mathbb{D}} q(f', f') dS, \quad P_q(f) = \frac{2}{\pi} \int_{\mathbb{D}} \log \frac{1}{|z|} q(f', f') dS.$$

Thus $[-R_k] = 2\pi c_1(K_X)$. Write $D > 0$ for the uniform bound on the central derivative of every holomorphic disc. This D is unrelated to the domain in Corollary 10.2.

[§2; Lemma 2.1](#) · pp. 4–5

1 Centered disc variations

$$J(f) = d(f) + P_q(f) - \delta A_h(f) \quad \mathcal{L}J \leq 0$$

Boundary regularity and frames give actual variations and trace identities

[Proposition 5.2 · pp. 17–18](#)

[Proposition 6.4; Lemma 6.3 · p. 21](#)

[Proposition 7.1 · p. 22](#)

2 Two outputs of the same Hessian

$$h = k, q = R_k: K_X \text{ nef. } q = -h, \delta = 1: R_h \geq -h \Rightarrow h^n \geq ck^n.$$

Solve in classes near the nef class; integrate the uniform lower bound

[Monge–Ampère existence \[MA\] · §1, equation \(1.1\) · print pp. 366–367 /](#)

[PDF pp. 3–4](#)

[Proposition 8.2 · pp. 23–24](#)

3 Join by Monge–Ampère

$$[h_t] = \alpha + t[k], R_{h_t} = -h_t + tk, \quad \int_X \alpha^n > 0$$

The negative-index error is $O(\varepsilon)$; use Morse and projectivity criteria

[Holomorphic Morse inequalities \[HM\] · \(0.7\); Theorem 0.8\(a\) · print pp.](#)

[193–194 / PDF pp. 6–7](#)

[Moishezon projectivity \[MP\] · Theorem 6 · p. 3](#)

4 Morse criterion, then projectivity

$$K_X \text{ big, } X \text{ Moishezon} + \text{Kähler} \Rightarrow X \text{ projective}$$

A big decomposition and klt perturbation reduce to nef plus ample

[Proposition 9.3 · pp. 25–26](#)

[Log cone and ampleness criteria \[LC\] · Theorem 1.1\(5\) · print pp.](#)

[728–729 / PDF pp. 2–3](#)

[Log cone and ampleness criteria \[LC\] · §4.4; Theorem 4.10 · print pp.](#)

[737–738 / PDF pp. 11–12](#)

5 No rational curves: ampleness and embedding

$$(1 + \eta m)K_X \sim_{\mathbb{Q}} (K_X + \eta E) + \eta H \quad \text{ample}$$

$$K_X^{\otimes r} \text{ gives a projective embedding for large } r$$

Theorem 1.1: joining the analytic outputs, then passing through projectivity to ampleness

Open figure ↗ TeX source ↓

1. Two uses of the same variation

Maximize $J = d + P_q - \delta A_h$ among finite-area discs with a fixed center. The area penalty gives attainment, and comparison with dilations supplies boundary regularity. Compute the Hessian along centered variations in a boundary- h -orthonormal holomorphic frame. The choice $h = k, q = R_k, \delta \downarrow 0$ bounds the Ricci integral of every test disc; $q = -h, \delta = 1$ gives a lower volume bound under $R_h \geq -h$.

[Proposition 3.2; Lemma 3.3 · pp. 6–8](#) · [Proposition 5.2 · pp. 17–18](#) · [Proposition 6.4; Lemma 6.3 · p. 21](#) · [Proposition 7.1 · p. 22](#)

2. Feed the nef class into the volume estimate

Put $\alpha = 2\pi c_1(K_X)$. Nefness makes $\alpha + t[k]$ a Kähler class for $t > 0$. The negative-sign Monge–Ampère equation supplies $[h_t] = \alpha + t[k]$ with $R_{h_t} = -h_t + tk$. Integrate the uniform lower bound and let $t \downarrow 0$ in cohomology to obtain $\int_X c_1(K_X)^n > 0$.

[Monge–Ampère existence \[MA\] · §1, equation \(1.1\) · print pp. 366–367 / PDF pp. 3–4](#) · [Proposition 8.2 · pp. 23–24](#)

3. Prepare the Morse criterion and obtain projectivity

For a nef line bundle L , put $\beta = 2\pi c_1(L)$ and choose $\Theta_\varepsilon \geq -\varepsilon k$. On the locus with a negative eigenvalue,

$$|\Theta_\varepsilon^n| \leq n\varepsilon(\Theta_\varepsilon + 2\varepsilon k)^{n-1} \wedge k.$$

The integral on the right is $O(\varepsilon)$ by cohomology. If $\int_X \beta^n > 0$, the integral on the nondegenerate locus of index at most one is positive as well. The Morse inequalities make L big and X Moishezon. Apply this with $L = K_X$, then use Namikawa’s projectivity theorem with the retained Kähler and smoothness assumptions. Projectivity was not an initial hypothesis.

[Lemma 9.1 · pp. 24–25](#) · [Holomorphic Morse inequalities \[HM\] · \(0.7\); Theorem 0.8\(a\) · print pp. 193–194 / PDF pp. 6–7](#) · [Theorem 9.2 · p. 25](#) · [Moishezon projectivity \[MP\] · Theorem 6 · p. 3](#)

4. From absence of rational curves to ampleness and an embedding

A nonconstant map $\mathbb{P}^1 \rightarrow X$ would contradict hyperbolicity after restriction to $\mathbb{C}$. Now X is smooth and projective and K_X is big, so the hypotheses of the criterion recorded by Diverio–Trapani hold. Proposition 9.3 proves that criterion again through the cone theorem as follows.

Choose a smooth very ample hyperplane section H . Compare the growth of sections from bigness with the $O(m^{n-1})$ bound on H from the weak Morse inequality, using the restriction exact sequence. This yields $mK_X \sim H + E$ for some $m > 0$ and effective E .

[Holomorphic Morse inequalities \[HM\] · Theorem 0.1\(a\) · print p. 189 / PDF p. 2](#)

Choose a sufficiently small positive rational η so that $(X, \eta E)$ is klt. Absence of rational curves and the log cone theorem make $N = K_X + \eta E$ nef. Hence

$$(1 + \eta m)K_X \sim_{\mathbb{Q}} N + \eta H$$

is ample. Clearing the positive rational coefficient makes K_X itself ample, and a sufficiently high positive tensor power gives the projective embedding.

Proposition 9.3 · pp. 25–26 · Canonical ampleness criterion [DT] · Lemma 2.1 · p. 3 ·
 Log cone and ampleness criteria [LC] · Theorem 1.1(5) · print pp. 728–729 / PDF pp. 2–3 ·
 Log cone and ampleness criteria [LC] · §4.4; Theorem 4.10 · print pp. 737–738 / PDF pp. 11–12 ·
 §9; completion of Theorem 1.1 · pp. 24–26

3. Proposition 5.2 — Boundary normalization and the Hessian

Proof target : Proposition 5.2 · pp. 17–18

Proposition 5.2 — Three trace identities

Use X, k as above and a Kähler metric h . The disc f and holomorphic frame $B = (b_1, \dots, b_n)$ are continuous and W^{1,p_0} on the closed disc, with $p_0 > 2$; B is everywhere nonsingular and satisfies $B^* h B = I$ on the boundary. The centered holomorphic family F_t satisfies $F_0 = f$, $F_t(0) = x = f(0)$ and $\partial_{t_j} F|_0 = z b_j$, and depends holomorphically on the parameters with values in local W^{1,p_0} coordinates. Put $\mathcal{L} = \sum_j \partial_{t_j} \partial_{\bar{t}_j} |_0$. For every smooth closed real $(1, 1)$ -form q ,

$$\mathcal{L}d = \text{tr}(B(0)^* k(x) B(0)), \quad \mathcal{L}P_q = \frac{1}{2\pi} \int_0^{2\pi} \text{tr}_h q(f(e^{i\theta})) d\theta, \quad \mathcal{L}A_h = n - A_{R_h}(f).$$

Proposition 5.2 · pp. 17–18

1 Area penalty and attainment

$$J = d + P_q - \delta A_h, \quad A_k(f) - A_{k,r}(f) \leq C_f(1 - r)$$

The dilation tail gives Sobolev regularity; normalize in a weighted Hardy space

[Proposition 3.4 · pp. 8–9](#)

[Lemma 4.1; Proposition 4.2 · pp. 10–12](#)

[Oka principle \[OK\] · Theorem 1.3; §1, example \(A\) · pp. 1–3](#)

2 A boundary-normalized holomorphic frame

$$f, B \in W^{1,p_0} \cap C^0, \quad p_0 > 2, \quad B^* h B = I \text{ on } \partial \mathbb{D}$$

Use a centered right inverse and the complex implicit function theorem

[Cauchy–Green operator \[CG\] · §4.1 · p. 11](#)

[Banach implicit functions \[IF\] · Theorem 2.3\(ii\), \(d\) · p. 11](#)

[Proposition 4.3 · pp. 13–16](#)

3 Realize centered variations

$$Hg = Tg - (Tg)(0), \quad g + Q(zSt + Hg) = 0$$

$$F_t(0) = x, \quad \partial_{t_j} F|_0 = zb_j$$

Move derivatives to the boundary; $N = I$ removes the trace–determinant error

[Proposition 5.2 · pp. 17–18](#)

[Lemma 5.1 · p. 17](#)

4 Three identities via the weak boundary limit

$$\mathcal{L}d = \operatorname{tr}(B(0)^* k B(0))$$

$$\mathcal{L}P_q = \frac{1}{2\pi} \int_0^{2\pi} \operatorname{tr}_h q(f(e^{i\theta})) d\theta, \quad \mathcal{L}A_h = n - A_{R_h}(f)$$

Proposition 5.2: from extremal discs to admissible variations and three identities

[Open figure ↗](#) [TeX source ↓](#)

1. Area penalty and the boundary tail

Lemma 3.1 carries out Brody reparametrization to bound the central derivative. On an outer annulus, $-\delta A_h$ dominates the logarithmic term; Fatou’s lemma and normal convergence in the interior give attainment. Comparing with $f(rz)$ yields $A_q(f) \geq -d(f)$ and $A_k(f) - A_{k,r}(f) \leq C_f(1 - r)$. Local mean-value estimates and summation over dyadic annuli give continuity up to the boundary and $W^{1,p}$ regularity for $2 < p < 4$. No uniformity of C_f over all metrics is asserted.

[Proposition 3.2; Lemma 3.3 · pp. 6–8](#) [Proposition 3.4 · pp. 8–9](#)

2. Realize the directions by genuine disc families

Lemma 4.1 uses the Cauchy–Green right inverse and the Oka principle to trivialize a holomorphic bundle over a slightly larger disc. Proposition 4.2 normalizes the frame in a weighted Hardy space; rotation-difference estimates for a Toeplitz operator then supply boundary W^{1,p_0} regularity and nonsingularity. Proposition 4.3 subtracts the central value from the right inverse, $Hg = Tg - (Tg)(0)$, and applies the complex Banach implicit function theorem to

$$g + Q(zSt + Hg) = 0, \quad Q(0) = DQ(0) = 0.$$

This preserves both the center and the prescribed first variation. The background on spectral factorization is not treated as automatically supplying the required boundary regularity.

Lemma 4.1; Proposition 4.2 · pp. 10–12 · Proposition 4.3 · pp. 13–16 ·

Cauchy–Green operator [CG] · §4.1 · p. 11 · Oka principle [OK] · Theorem 1.3; §1, example (A) · pp. 1–3 ·
Banach implicit functions [IF] · Theorem 2.3(ii), (d) · p. 11

3. Pass through the weak boundary limit

The inequality $p_0/2 > 1$ justifies parameter differentiation of the area and Poisson integrals. Closedness gives the local identity $\mathcal{L}q(F_z, F_z) = \partial_z \partial_{\bar{z}}(|z|^2 \operatorname{tr}(B^* q B))$. For $N = B^* h B$, the boundary value $N = I$ allows replacement of $\partial_r \operatorname{tr} N$ by $\partial_r \log \det N$ along suitable radii. The error vanishes by a radial $W^{1,2}$ estimate. Green’s formula gives the three identities and $\log \det N(0) = P_{R_h}(f)$; maximality then gives $\mathcal{L}J \leq 0$. No holomorphic extension across the boundary is used.

Proposition 5.2 · pp. 17–18 · Lemma 5.1 · p. 17 · Corollary 5.3 · p. 19

4. Proposition 6.4 — Nefness of the canonical class

Proof target : Proposition 6.4; Lemma 6.3 · p. 21

Proposition 6.4 — Arbitrarily small negative curvature error

Under the hypotheses of Theorem 1.1 and for the fixed Kähler metric k , for every $\varepsilon > 0$ there is a smooth real function u_ε satisfying

$$-R_k + i\partial\bar{\partial}u_\varepsilon \geq -\varepsilon k.$$

Consequently $2\pi c_1(K_X)$ is nef.

Proposition 6.4; Lemma 6.3 · p. 21

1 A bound for each fixed test disc

$h = k, q = R_k, P_{R_k}(g) \leq C$ for every finite-area disc g

Compare before taking $\delta \downarrow 0$; exclude the envelope value $-\infty$

[Proposition 6.1 · pp. 19–20](#)

[Poletsky–Rosay envelopes \[PR\] · Theorem 1.1 · print p. 1407 / PDF p. 1](#)

2 A psh envelope on the determinant bundle

$Y = (\det T^{1,0} X) \setminus 0_X, \ell = \log |s|_k^2$

$\ell - C \leq v \leq \ell, \quad v(\lambda s) = v(s) + \log |\lambda|^2$

Descend the weights by homogeneity, keeping the canonical-bundle sign

[Proposition 6.2 · pp. 20–21](#)

3 Read the curvature of the dual bundle

$v - \ell = u \circ \pi, \quad -C \leq u \leq 0$

$|s_i^*|^2 = e^{-(a_i + u)}, \quad -R_k + i\partial\bar{\partial}u \geq 0$

Control overlap errors of local suprema; glue by cutoffs and regularized maxima

[Appendix A; Lemma A.1 · pp. 27–30](#)

[Proposition 6.4; Lemma 6.3 · p. 21](#)

4 Smooth the bounded potential

$\forall \varepsilon > 0 \quad \exists u_\varepsilon \in C^\infty(X, \mathbb{R}) : \quad -R_k + i\partial\bar{\partial}u_\varepsilon \geq -\varepsilon k$

Proposition 6.4: the bound on all discs, the determinant-bundle envelope, and smoothing

[Open figure ↗](#) [TeX source ↓](#)

1. From maximizers to every test disc

Take $h = k, q = R_k$ and maximize for $0 < \delta \leq 1$. Corollary 5.3 and $A_{R_k}(f) \geq -D$ imply $\operatorname{tr} N(0) \leq n + D + M$, where $M = \sup_X |\operatorname{tr}_k R_k|$. The determinant identity and arithmetic–geometric mean bound $P_{R_k}(f)$ uniformly from above. Fix an arbitrary finite-area disc g , compare it with the maximum, and only then let $\delta \downarrow 0$. This gives $P_{R_k}(g) \leq C$ without a uniform area bound on the maximizing discs in this limit.

[Proposition 6.1 · pp. 19–20](#) [Corollary 5.3 · p. 19](#)

2. Turn envelope homogeneity into a canonical metric

On $Y = (\det T^{1,0} X) \setminus 0_X$, put $\ell(s) = \log |s|_k^2$ and let v be its Poisson envelope. The preceding estimate on projected discs gives $\ell - C \leq v \leq \ell$, excluding the identically $-\infty$ alternative in the Poletsky–Rosay theorem. Homogeneity $v(\lambda s) = v(s) + \log |\lambda|^2$ gives $v - \ell = u \circ \pi$ with $-C \leq u \leq 0$. For local frames s_i and $a_i = \log |s_i|_k^2$, the weights $a_i + u$ are bounded and psh. Setting $|s_i^*|^2 = e^{-(a_i+u)}$ on the dual frames yields curvature $-R_k + i\partial\bar{\partial}u \geq 0$ on K_X .

Proposition 6.2 · pp. 20–21 · Poletsky–Rosay envelopes [PR] · Theorem 1.1 · print p. 1407 / PDF p. 1

3. Smooth without assuming continuity

Lemma 6.3 supplies smoothing of the bounded potential. Appendix A compares local suprema of psh potentials at fixed multiples of a radius, bounding overlap errors by $O(1/|\log b|) + O(b)$. Choose a radius making this error small and glue by a regularized maximum after cutoff corrections. The curvature loss can be made arbitrarily small. Applying this to Proposition 6.2 proves the claim. The alternative citation to Demailly’s regularization theorem mentioned in the manuscript is not additionally checked here.

Proposition 6.4; Lemma 6.3 · p. 21 · Appendix A; Lemma A.1 · pp. 27–30

5. Proposition 7.1 — A uniform volume lower bound

Proof target : Proposition 7.1 · p. 22

Proposition 7.1 — A constant common to all eligible metrics

Under the hypotheses of Theorem 1.1, fix k and let D be the derivative bound of Lemma 3.1. Every smooth Kähler form h satisfying $R_h \geq -h$ obeys

$$h^n \geq c_{n,D} k^n, \quad c_{n,D} = e^{-D} \left(\frac{n}{2n+D} \right)^n > 0.$$

Proposition 7.1 · p. 22

1 Compare with a constant disc

$$q = -h, \delta = 1, R_h \geq -h$$

$$P_h(f) + A_h(f) \leq d(f) \leq D$$

Insert the Ricci bound and maximality into the trace identities

[Corollary 5.3 · p. 19](#)

[Lemma 5.1 · p. 17](#)

2 Determinant below, trace above

$$\det(B(0)^* h B(0)) = e^{P_{R_h}(f)} \geq e^{-D}$$

$$\operatorname{tr}(B(0)^* k B(0)) \leq 2n + D$$

Bound the denominator by arithmetic-geometric mean; the constant is uniform

[Proposition 7.1 · p. 22](#)

3 Cancel the frame in the ratio

$$\frac{h^n}{k^n} = \frac{\det(B(0)^* h B(0))}{\det(B(0)^* k B(0))} \geq e^{-D} \left(\frac{n}{2n + D} \right)^n$$

Proposition 7.1: divide a determinant lower bound by a trace-controlled determinant

[Open figure ↗](#) [TeX source ↓](#)

1. Compare with the constant disc

At each point choose a maximizer with $q = -h, \delta = 1$. Comparison with the constant disc, whose value is zero, gives $P_h(f) + A_h(f) \leq d(f) \leq D$. The curvature hypothesis implies

$P_{R_h}(f), A_{R_h}(f) \geq -D$. Insert $\operatorname{tr}_h(-h) = -n$ into the maximizing inequality to obtain $\operatorname{tr}(B(0)^* k B(0)) \leq 2n + D$.

[Proposition 7.1 · p. 22](#) · [Corollary 5.3 · p. 19](#)

2. Cancel the frame determinant

Lemma 5.1 gives $\det(B(0)^* h B(0)) = e^{P_{R_h}(f)} \geq e^{-D}$. The preceding trace bound controls the denominator determinant by arithmetic-geometric mean. Taking their ratio cancels $|\det B(0)|^2$ and gives

$$\frac{h^n}{k^n} = \frac{\det(B(0)^* h B(0))}{\det(B(0)^* k B(0))} \geq e^{-D} \left(\frac{n}{2n + D} \right)^n.$$

The auxiliary disc, frame and size of the variation may depend on h and the point. None appears in the final constant.

[Lemma 5.1 · p. 17](#) · [Proposition 7.1 · p. 22](#)

6. Proposition 8.2 — Positive top self-intersection

Proof target : [Proposition 8.2 · pp. 23–24](#)

Proposition 8.2 — Positive canonical volume

Under the hypotheses of Theorem 1.1, let X be a compact connected Kähler manifold of positive dimension n with no nonconstant holomorphic map $\mathbb{C} \rightarrow X$. Then

$$\int_X c_1(K_X)^n > 0.$$

[Proposition 8.2 · pp. 23–24](#)

1 Kähler classes near the nef class

$$\alpha = [-R_k], \quad \gamma_t = -R_k + tk + i\partial\bar{\partial}a_t > 0$$

Use nefness to choose γ_t ; prescribe $e^{a_t}k^n$ in the existence theorem

[Proposition 6.4; Lemma 6.3 · p. 21](#)

[Monge–Ampère existence \[MA\] · §1, equation \(1.1\) · print pp. 366–367 /](#)

[PDF pp. 3–4](#)

2 Solve for an arbitrary positive volume form

$$h_t = \gamma_t + i\partial\bar{\partial}w_t, \quad h_t^n = e^{a_t + w_t}k^n$$

$$R_{h_t} = -h_t + tk \geq -h_t$$

Integrate the uniform bound and take $t \downarrow 0$ without metric convergence

[Proposition 7.1 · p. 22](#)

[Proposition 8.2 · pp. 23–24](#)

3 Evaluate the intersection polynomial at zero

$$\int_X (\alpha + t[k])^n \geq c_{n,D} \int_X k^n$$

$$(2\pi)^n \int_X c_1(K_X)^n \geq c_{n,D} \int_X k^n > 0$$

Proposition 8.2: choose Kähler classes and transfer the volume bound to cohomology

[Open figure ↗](#) [TeX source ↓](#)

1. Existence for an arbitrary positive volume form

Put $\alpha = [-R_k]$ and use nefness to choose $-R_k + i\partial\bar{\partial}a_t \geq -(t/2)k$. Apply Theorem 8.1 to $\gamma_t = -R_k + tk + i\partial\bar{\partial}a_t > 0$ to obtain

$$h_t = \gamma_t + i\partial\bar{\partial}w_t > 0, \quad h_t^n = e^{a_t + w_t}k^n.$$

The prescribed volume form is $e^{a_t}k^n$; no prior normalization of its total volume is needed because the additive constant of the solution enters the right-hand side. Logarithmic differentiation gives $R_{h_t} = -h_t + tk \geq -h_t$.

[Theorem 8.1 · p. 23](#) [Monge–Ampère existence \[MA\] · §1, equation \(1.1\) · print pp. 366–367 / PDF pp. 3–4](#) [Proposition 8.2 · pp. 23–24](#)

2. Take the limit of the intersection polynomial

Apply Proposition 7.1 with the same constant for every $t > 0$. Since $[h_t] = \alpha + t[k]$,

$$\int_X (\alpha + t[k])^n = \int_X h_t^n \geq c_{n,D} \int_X k^n > 0.$$

The left-hand side is a polynomial in t . Evaluation at zero gives

$(2\pi)^n \int_X c_1(K_X)^n \geq c_{n,D} \int_X k^n > 0$. No convergence of h_t or its potentials is assumed in this step.

[Proposition 7.1 · p. 22](#) · [Proposition 8.2 · pp. 23–24](#)

7. Corollary 10.1 — All exponents from the freeness input in 038

Proof target : [Corollary 10.1 · p. 26](#)

1 First obtain projectivity and ampleness

X smooth projective, $L = K_X$ ample

Apply the all-exponents corollary in 038 with $L = K_X$

[Theorem 1.1 · p. 1](#)

[Fujita freeness \[FF\] · Corollary 6.3 · p. 23](#)

[Corollary 10.1 · p. 26](#)

2 Shift every eligible exponent by one

$$m \geq n + 2 \implies m - 1 \geq n + 1$$

$$K_X \otimes L^{\otimes(m-1)} = K_X^{\otimes m} \text{ globally generated}$$

Corollary 10.1: obtain projectivity and ampleness, then shift the exponent by one

[Open figure ↗](#) [TeX source ↓](#)

1. Meet the hypotheses of the input

Theorem 1.1 makes X a smooth connected projective variety with K_X ample. Hence the Fujita freeness input from catalogue 038 applies with $L = K_X$. This input is not used retroactively in the proof of the main theorem.

[Theorem 1.1 · p. 1](#) · [Fujita freeness \[FF\] · Corollary 6.3 · p. 23](#)

2. Preserve the quantifier over all integers

For every integer $m \geq n + 2$, one has $m - 1 \geq n + 1$. Corollary 6.3 of the input therefore makes

$$K_X \otimes L^{\otimes(m-1)} = K_X^{\otimes m}$$

globally generated. The all-exponents corollary, rather than only the smallest-exponent main theorem, is the required input. The [article on 038](#) describes that extension.

[Corollary 10.1 · p. 26](#) · [Fujita freeness \[FF\]](#) · [Corollary 6.3 · p. 23](#)

8. Corollary 10.2 — From classification to a finite cover

Proof target : [Corollary 10.2 · pp. 26–27](#)



Corollary 10.2: semialgebraic-cover classification and a finite-index deck subgroup

[Open figure ↗](#) [TeX source ↓](#)

1. Remove the extra factor in the classification

First use Theorem 1.1 to make X projective, then apply Theorem 1.1 of 058 to its ordinary universal cover. The resulting decomposition is $\tilde{X} \simeq D \times \mathbb{C}^m \times F$, with F simply connected, normal and projective. Smoothness of the product forces F to be smooth. Hyperbolicity passes to covers and factors, so $m = 0$ and F is Brody hyperbolic. The proof of the classification theorem itself is treated as an external input.

Corollary 10.2 · pp. 26–27 · Semialgebraic universal covers [SC] · Theorem 1.1 · p. 1

2. Use finite automorphisms to obtain a product action

If $\dim F > 0$, reapply Theorem 1.1 to F to make K_F ample. Thus $c_1(F) < 0$, and Kobayashi's theorem makes $\mathbf{Aut}(F)$ finite. Finiteness is immediate when F is a point. The D -component of a deck transformation is constant on compact connected F ; applying the same observation to the inverse gives an automorphism of the base. The remaining F -component is a family in a finite group over connected D , hence independent of the base. The argument includes the case where D is a point.

Corollary 10.2 · pp. 26–27 · Finite automorphisms [KA] · Theorem · print p. 184 / PDF p. 1

3. Check the action of the finite-index kernel

Let Γ_0 be the kernel of the homomorphism from the deck group Γ to $\mathbf{Aut}(F)$. It has finite index and acts faithfully and freely on D . For compact $K \subset D$, the set $K \times F$ is compact, so proper discontinuity of the deck action descends to D , and Γ_0 is discrete. The quotient is $(D/\Gamma_0) \times F$. Its map to X is a connected finite unramified cover, hence a finite étale cover of the projective variety. Compactness of this cover makes D/Γ_0 compact, proving cocompactness as well.

Corollary 10.2 · pp. 26–27

9. Which papers supply which steps

Direct inputs split into those supporting the main analytic and positivity argument and those used only in its corollaries. Diverio–Trapani supplies a comparison statement: the manuscript proves Proposition 9.3 again through the cone theorem. The checks below concern statements and application hypotheses, not complete external proofs.

Source

Oka principle, Cauchy–Green operator, Banach implicit functions

Use and role

Lemma 4.1 and Proposition 4.3: the frame bundle over a Stein disc, centered right inverse, invertible linearization

Checking scope

Specified results and the bundle/function-space hypotheses compared

Source

Poletsky–Rosay envelopes

Use and role

Proposition 6.2: a continuous weight on a connected manifold and exclusion of $-\infty$

Checking scope

Theorem 1.1 and the envelope bounds compared

Source

Monge–Ampère existence

Use and role

Proposition 8.2: a Kähler class and an arbitrary positive volume form

Checking scope

Existence statement at Berman’s equation (1.1) and the sign convention compared

Source

Holomorphic Morse inequalities, Moishezon projectivity

Use and role

Lemma 9.1 and Theorem 9.2: positive index integral, smooth Kähler space

Checking scope

Hypotheses and the positivity connection compared

Source

Canonical ampleness criterion, Log cone and ampleness criteria

Use and role

Proposition 9.3: projective, big, no rational curves, klt perturbation

Checking scope

Criterion, cone and Kleiman statements compared; original resolution theorem not checked

Source

Fujita freeness (038)

Use and role

Only Corollary 10.1: the condition on every integer exponent

Checking scope

Corollary 6.3 reread in the source; its proof not reverified

Source

Semialgebraic universal covers (058), Finite automorphisms

Use and role

Only Corollary 10.2: product classification and passage to finite index

Checking scope

Classification statement and hypotheses of the finiteness theorem compared

Oka principle [OK] · Theorem 1.3; §1, example (A) · pp. 1–3 · Cauchy–Green operator [CG] · §4.1 · p. 11 ·
Banach implicit functions [IF] · Theorem 2.3(ii), (d) · p. 11 ·
Poletsky–Rosay envelopes [PR] · Theorem 1.1 · print p. 1407 / PDF p. 1 ·
Monge–Ampère existence [MA] · §1, equation (1.1) · print pp. 366–367 / PDF pp. 3–4 ·
Holomorphic Morse inequalities [HM] · (0.7); Theorem 0.8(a) · print pp. 193–194 / PDF pp. 6–7 ·
Moishezon projectivity [MP] · Theorem 6 · p. 3 · Canonical ampleness criterion [DT] · Lemma 2.1 · p. 3 ·
Log cone and ampleness criteria [LC] · Theorem 1.1(5) · print pp. 728–729 / PDF pp. 2–3 ·
Log cone and ampleness criteria [LC] · §4.4; Theorem 4.10 · print pp. 737–738 / PDF pp. 11–12 ·
Fujita freeness [FF] · Corollary 6.3 · p. 23 · Semialgebraic universal covers [SC] · Theorem 1.1 · p. 1 ·
Finite automorphisms [KA] · Theorem · print p. 184 / PDF p. 1

10. Return to the sources

- Proposition 3.2; Lemma 3.3 · pp. 6–8 — Attainment and the boundary-area tail.
- Lemma 4.1; Proposition 4.2 · pp. 10–12 — Boundary normalization and regularity.
- Proposition 4.3 · pp. 13–16 — Realizing formal directions by holomorphic families.
- Proposition 5.2 · pp. 17–18 — Differentiation under the integrals and the weak boundary limit.
- Proposition 6.2 · pp. 20–21 — The determinant-bundle envelope and the canonical-bundle sign.
- Appendix A; Lemma A.1 · pp. 27–30 — Gluing bounded, possibly discontinuous potentials.
- Proposition 7.1 · p. 22 — The lower bound independent of auxiliary choices.
- Proposition 8.2 · pp. 23–24 — The cohomological limit without metric convergence.
- §9; completion of Theorem 1.1 · pp. 24–26 — The inputs from bigness to projectivity and ampleness.
- Corollary 10.1 · p. 26 — Use of the all-powers input from 038.
- Corollary 10.2 · pp. 26–27 — The classification from 058 and the finite-index subgroup.

The principal reductions, the frame/variation/Hessian connections in §§3–5, and the relevant passages of §§6–10 and Appendix A were read. The specified results and application hypotheses in the twelve listed external works were compared. Every conclusion of the main theorem and its two corollaries is linked to a diagram and the explanation immediately following it.

This does not independently re-prove all functional-analytic estimates for boundary frames, Sobolev substitution, the weak boundary limit, or all local estimates in Appendix A. The original resolution theorem of Kollár cited in Proposition 9.3, the alternative route through Demailly’s regularization theorem, and all methodological background sources were not checked here. Complete external proofs, especially freeness in 038 and classification in 058, are outside this article’s checking scope. The account is neither an exhaustive dependency audit nor expert peer review or formal verification.